STUDENT ASSIGNMENT: BUILD A SMART SEARCH & DOCUMENT Q&A; SYSTEM

A step-by-step beginner project to build a smart search and answer system using document reading, local database matching, and AI generation.

Course: Introduction to Artificial Intelligence	**Due Date:** August 28, 2026 @ 23:59 UTC
Project: Smart Document Q&A; App	**Weight:** 25% of final grade (Individual or Teams of 2)

1. Project Goal

Your goal is to build a simple program that reads uploaded document files, searches their contents, and uses a local AI to answer user questions based on those documents. If the answer is not in the text, the AI must reply 'I don't know' instead of guessing.

2. Key Requirements

- **Document Ingestion:** Read text from uploaded files (PDF, Word, or Text) and divide them into smaller segments (chunks) so they are easy to search.
- **Database Storage:** Store the text chunks in a database so they can be matched with user queries.
- **Retrieval & Sorting:** Search the database for relevant chunks and sort them to find the best match.
- **AI Generation:** Ask a local AI model to answer the user's question using only the matching text context.
- **Safety Rule:** If the text doesn't contain the answer, the AI must reply 'I don't know'.

3. How to Build It (Step-by-Step)

1. **Setup:** Create a basic application workspace and connect it to a database collection.
2. **Chop Text:** Write code to open documents, extract their text, and slice them into search chunks.
3. **Search:** Convert chunks to numbers, save them in the database, and write a function to search for matches.
4. **Re-rank:** Use a sorting model to find the best matching chunk and retrieve its text.
5. **Ask AI:** Write the AI prompt with the safety rules, and send the request to a local AI server to get the final answer.

4. Grading Rubric

Component	Weight	What we look for
Document Processing	25%	Files load correctly and are split into chunks for search.
Search & Sorting	25%	The database search returns the correct chunks and the sorting model is active
Web API & Interface	20%	The user can upload files and ask questions using simple web endpoints.
AI Answers & Safety	15%	The AI answers using only the text, and says 'I don't know' for missing answers
Code & Documentation	15%	The code is clean, well-commented, and includes a setup guide (README).

5. What to Turn In

Submit your Python code file, a **requirements.txt** showing the libraries used, and a **README.md** file explaining how to run the project.